



Research paper

Selective electron transport at CsPbI₃(001)/TiO₂(001) interfaces: Finite-temperature electronic structure insights

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ABSTRACT

Accurate descriptions of the electronic structure of halide perovskites at finite temperatures require structural models that capture the dynamical tilting of BX₆ octahedra. Here, we demonstrate that a single static structure—constructed by combining the low-temperature γ -phase geometry with the cubic lattice of the α -phase—reproduces results obtained from time-averaged trajectories. Using this approach, we investigate how free surfaces and interfaces modify the electronic structure of lead-halide perovskites. Focusing on CsPbI₃ as a model halide perovskite interfaced with the prototypical electron transport layer TiO₂, we make quantitative predictions of band offsets and band bending for the two possible CsPbI₃(001) surface terminations, PbI₂ and CsI. Our results reveal that TiO₂ acts as an efficient electron transport layer only in the case of PbI₂ termination, highlighting the termination-dependent selectivity of charge transport at halide perovskite/TiO₂ interfaces.

1. Introduction

Despite long-standing challenges such as insufficient long-term stability and the presence of toxic Pb, [1–3] halide perovskites remain among the most promising and intensively studied absorber materials for next-generation solar cells [4,5]. The highest-performing perovskite absorbers are still Pb-based ABX₃ compounds, typically incorporating mixed halides and suitable monovalent cations on the X and A sites, respectively [6].

With the formulation of absorber compositions converging [6,7], remaining progress will depend on addressing Pb toxicity through encapsulation strategies and enhancing stability by tuning A- and X-site chemistry. Further breakthroughs in both efficiency and stability are expected to come from optimizing the interfaces between absorbers and contact layers, as well as the overall device architecture [8].

Interface engineering has emerged as a critical factor in determining the performance of perovskite solar cells, with extensive computational and experimental investigations revealing the complex interplay between structural, electronic, and chemical phenomena at hetero-interfaces [8]. Recent first-principles studies have highlighted the importance of properly modeling these interfaces. For instance, Haruyama et al. [9] investigated electron injection and defects at the TiO₂/CH₃NH₃PbI₃ (MAPI) interface, demonstrating how interfacial defects and charge transfer dynamics critically influence device performance. Their work underscored the need for accurate atomistic models

that capture both geometric and electronic properties at perovskite-oxide interfaces.

Beyond idealized crystalline interfaces, the role of structural disorder in mixed-cation perovskites has gained significant attention. Coppola et al. [10] employed first-principles calculations to characterize the electronic structure and interfacial properties of triphenylamine- and phenothiazine-based hole transport materials with MAPI, considering different surface terminations and their impact on hole extraction. More recently, Pecoraro et al. [11] addressed charge dynamics at the interface between disordered mixed-cation perovskites and the widely used spiro-MeOTAD hole transport layer, revealing how local compositional fluctuations and cation orientations affect interfacial charge transfer rates. These studies demonstrate that capturing realistic structural features—including cation disorder and surface termination effects—is essential for predicting interfacial electronic properties.

Precise electronic-structure calculations play a vital role in advancing these technologies. Modern density functional theory (DFT) methods now provide quantitatively accurate descriptions of halide perovskites [12] as well as contact materials—typically transparent conductive oxides (TCOs) used as charge transport layers—at a reasonable computational cost [7,13]. Nevertheless, such calculations must carefully account for several physical phenomena: electron correlation effects in both material classes [14,15], spin-orbit coupling (SOC) arising from the heavy Pb atoms in lead-based absorbers [16], and realistic structural models capturing finite-temperature behavior [17].

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Although electron correlation and SOC are now routinely incorporated into DFT workflows with minimal computational overhead, representing finite-temperature structures has traditionally necessitated expensive molecular dynamics (MD) simulations to generate thermal ensembles [12,17].

The computational cost of *ab initio* molecular dynamics (AIMD) becomes particularly prohibitive when studying complex systems such as surfaces and interfaces, where large supercells and extended sampling are required to achieve statistical convergence. While previous approaches have relied on averaging electronic properties over multiple MD snapshots [12,17] or approximating dynamic disorder through superpositions of differently oriented low-temperature polymorphs [18], we demonstrate here that a single, carefully constructed static structure can reproduce the time-averaged electronic properties of finite-temperature trajectories with high fidelity. Specifically, we show that the γ' phase—obtained by optimizing the low-temperature α -phase—captures the essential physics of octahedral tilting without requiring costly trajectory averaging. This computationally efficient approach enables systematic investigations of perovskite interfaces that would be prohibitively expensive using full AIMD-based workflows, while maintaining the accuracy necessary for quantitative predictions of band offsets and charge transport properties.

Having established an efficient methodology for finite-temperature electronic structure calculations, we extend this framework to investigate how free surfaces and interfaces modify the electronic properties of lead-halide perovskites relative to the bulk. Surfaces and interfaces introduce additional complexity through altered coordination environments, charge redistribution, and interfacial strain—all of which interact non-additively with temperature-induced octahedral distortions. Understanding these coupled effects is crucial for predicting band alignment and charge transport at realistic device interfaces.

In this work, we employ state-of-the-art DFT to investigate CsPbI_3 || TiO_2 as a model interface between a Pb-based perovskite absorber and a prototypical electron-transporting TCO. We first demonstrate that finite-temperature properties of Pb-based perovskites can be accurately reproduced using a static structural model, avoiding the need for costly trajectory sampling. We then examine how this structural model is modified at the perovskite || TCO interface compared to bulk materials. Finally, we present quantitative predictions of band offsets and band bending at CsPbI_3 || TiO_2 interfaces, highlighting a pronounced termination dependence: for PbI_2 -terminated surfaces, the crystalline interface enables selective electron transport without requiring an additional charge-extraction layer, whereas for CsI termination, valence-band alignment prevents such selectivity. By combining efficient static structural models with detailed interface analysis, this work provides both a practical computational framework and fundamental insights into termination-dependent charge transport mechanisms at halide perovskite-oxide interfaces.

2. Computational details

First-principles atomistic and electronic-structure calculations were performed with the QuantumATK package [19] using its DFT-LCAO (linear combination of atomic orbitals) [20] engine. A localized atomic-orbital basis set with optimized basis functions (converged for total energy and forces) was employed together with norm-conserving pseudopotentials. Reciprocal-space integrations used converged Monkhorst-Pack k-point meshes and an appropriate electronic smearing scheme for metallic/semiconducting systems.

Exchange-correlation effects were treated with the semi-local Perdew-Burke-Ernzerhof (PBE) functional [21], and long-range dispersion interactions were included during geometry optimizations via the Tkatchenko-Scheffler (TS) correction [22]. Geometry relaxations were carried out using the LBFGS optimizer with standard force and stress convergence criteria.

To reduce the self-interaction error and obtain improved band gaps at modest cost, the DFT-1/2 (DFT+1/2) correction [15] as implemented in QuantumATK was applied; element-specific half-occupation parameters were tuned following the established DFT-1/2 workflow [12]. Spin-orbit coupling (SOC) was included in all final electronic structure calculations where heavy-element relativistic effects are important. Specifically, SOC was applied in all band structure, density of states, and band offset calculations for systems containing Pb atoms, while it was omitted in geometry optimizations to reduce computational cost. After parameterization, DFT-1/2+SOC yields band gaps comparable to higher-level many-body approaches while incurring negligible additional runtime relative to the corresponding plain DFT+SOC calculations performed with the same DFT-LCAO setup.

2.1. Bulk structures

Crystalline CsPbI_3 was modeled in the cubic α -phase (Pm-3m), the orthorhombic γ -phase (Pbnm), and a pseudo-cubic γ' variant obtained by optimizing the γ -phase basis structure within the cubic α -phase lattice. The γ' structure retains cubic lattice vectors but includes tilted Pb-I octahedra (Pnma symmetry), which we have previously shown [9,23] to represent room-temperature (RT) CsPbI_3 more accurately than the static α -phase, commonly used in prior studies [18,24,25].

All geometries were optimized with a plane-wave cutoff of 550 eV. Convergence thresholds were 1×10^{-6} eV for total energies and 5×10^{-3} eV/Å for forces. Brillouin-zone integrations employed $3 \times 3 \times 3$ k-point grids for $(2 \times 2 \times 2)$ supercells of the primitive CsPbI_3 cell. For TiO_2 , rutile-phase structures ($\text{P4}_2/\text{mmn}$) were calculated with a comparable setup, using $6 \times 6 \times 8$ k-point grids.

2.2. Lattice matching and interfaces

Interface lattice constants were determined from energy-area curves, obtained by applying uniform in-plane strain (ϵ_{xy}) while fixing the out-of-plane lattice constant (z). Optimal strain values (ϵ_{xy} , ϵ_z) were identified by minimizing the combined energy of both strained materials. Although real interfaces may involve growth order, thickness effects, or reconstructions, this procedure provides thermodynamically consistent lattice parameters.

Slab thickness convergence was systematically tested for both CsPbI_3 and TiO_2 . For $\text{CsPbI}_3(001)$ surfaces, we examined slab thicknesses ranging from 2.5 to 4.5 formula units (f.u.), corresponding to 5 to 9 atomic layers. The band gap and band edge positions converged to within 0.1 eV for slabs with ≥ 3.5 f.u., which we adopted as our standard thickness. For $\text{TiO}_2(001)$ surfaces, slabs of 2.5 to 4.5 f.u. were tested, with convergence achieved at 3.5 f.u. The vacuum spacing of 25 Å was verified to eliminate spurious interactions between periodic images by comparing with calculations using 30 Å vacuum, which yielded energy differences below 0.01 eV.

Slab models of $\text{CsPbI}_3(001)$ surfaces (PbI_2 -terminated) were constructed with a thickness of 3.5 formula units (seven atomic layers), separated by 25 Å of vacuum. Interface structures for $\text{CsPbI}_3(001)$ || $\text{TiO}_2(001)$ were built analogously, considering both PbI_2 and CsI terminations. Interface relaxation allowed for atomic displacement along z , while in-plane lattice constants were fixed to optimized values. During interface relaxation, atoms within the central two layers of each material slab were held fixed at their optimized bulk positions to maintain bulk-like character in the interior, while all atoms within two layers of the interface were allowed to relax freely. Convergence criteria for interface relaxations were identical to those used for bulk structures.

Relative lateral alignments of $\text{TiO}_2(001)$ with PbI_2 - and CsI-terminated $\text{CsPbI}_3(001)$ were tested for primitive α -phase cells. For PbI_2 termination, the most stable configuration places I atop Ti sites, with Pb bridging two O atoms, outperforming shifts along (0,1/4), (0,1/2), (1/4,1/4), and (1/4,1/2) by at least 0.15 eV per interfacial formula unit. For CsI termination, a shift by (1/2,0) relative to the PbI_2 alignment